

Day – 9 Introduction to network protocols theory.

Task – 1 Wireshark to capture network traffic.

- 1. Two DNS Packets (Filter: dns)
- When your browser first attempts to visit Spotify, it needs to resolve the domain name into an IP address:
- DNS Packet 1 (Standard Query):
- Info Column: Standard query 0x... A www.spotify.com
- Details: This is the outgoing request sent from your computer to your configured DNS server (like 8.8.8.8 or your router) asking, "What is the IP address for www.spotify.com?"
- DNS Packet 2 (Standard Response):
- Info Column: Standard query response 0x... A www.spotify.com CNAME ...
- Details: This is the incoming reply from your DNS server providing the resolved canonical name (CNAME) and public IP addresses (often pointing to CDN edge nodes hosted by providers like Fastly or Akamai).
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- 2. Two HTTPS / TLS Packets (Filter: tls or tcp.port == 443)
- Once the IP address is resolved, your browser initiates a secure connection to Spotify's servers:
- TLS Packet 1 (Client Hello):
- Info Column: Client Hello
- Details: The very first packet of the TLS handshake sent from your browser. Inside the packet details under the Server Name Indication (SNI) extension, you will explicitly see www.spotify.com unencrypted, telling the server which website you want to load.
- TLS Packet 2 (Server Hello / Certificate):
- Info Column: Server Hello, Certificate, Server Key Exchange
- Details: The server's response acknowledging the handshake, presenting its SSL/TLS security certificate issued to Spotify, and agreeing on cryptographic parameters to establish the encrypted session.

Task – 2 Simulate the DNS desktop.

- Running a simulated nslookup command for www.flipkart.com in a terminal yields a response pointing to Flipkart's infrastructure:
- Plaintext
- Server: UnKnown
- Address: 192.168.1.1
- Non-authoritative answer:
- Name: www.flipkart.com
- Address: 163.53.76.86

Task – 3 Difference between HTTP and HTTPS.

- 1. Security and Encryption
- HTTP (HyperText Transfer Protocol): Data is transmitted in plain text. Anyone monitoring the network (like an attacker on public Wi-Fi) can easily intercept and read usernames, passwords, and sensitive information.
- HTTPS (HTTP Secure): Data is encrypted using TLS (Transport Layer Security). Even if someone intercepts the traffic, they only see unreadable scrambled code.
- 2. Default Network Ports
- HTTP: Communicates over port 80 by default.
- HTTPS: Communicates over port 443 by default, which is specifically routed through security layers.
- 3. Browser Trust and Indicators
- HTTP: Modern web browsers flag HTTP sites with a "Not Secure" warning or a broken icon in the address bar.
- HTTPS: Browsers display a padlock icon next to the URL, verifying that your connection to the server is authentic and private.

Task – 4 Based on scenario.

- However, trying this exercise with a brand-new website is one of the best ways to see the Domain Name System (DNS) in action. When you type a fresh URL into your browser, your local machine has no cached memory of its IP address, forcing a complete query sequence.
- Here is the exact sequence of DNS queries and responses you will observe in Wireshark when you filter for dns:

- 1. The Outgoing DNS Request (Standard query)
- What you see in the Info column: Standard query 0x... A <website-name> (and often a separate packet for AAAA for IPv6).
- What it means: Your computer sends a UDP packet (typically to port 53) directed at your configured DNS server (such as your home router, 8.8.8.8, or your ISP's resolver).
- 2. The Incoming DNS Response (Standard query response)
- What you see in the Info column: Standard query response 0x... A <website-name> A <IP-address>
- What it means: The DNS server replies back to your machine with the resolved answer. If you expand the packet details in Wireshark and look at the Answers section, you will see the returned IP address(es) and Time-To-Live (TTL) values, which tell your computer how long it is allowed to cache that record before querying again.

Task – 5 One protocol and it scenario.

- Protocol: DNS (Domain Name System)
- Real-World Scenario: Opening Spotify ([www.spotify.com])(https://www.spotify.com))
- Imagine you tap the Spotify icon on your phone to play your favorite playlist.
- The Problem: Your phone knows what "Spotify" is conceptually, but the underlying network hardware only understands numerical addresses (like 151.101.65.140). It cannot send a network packet to a text string.
- The Role of DNS: The moment you open the app, before a single note of music can stream, your phone quietly sends out a DNS request to a resolver (such as your home router or Cloudflare's 1.1.1.1).
- The Resolution: The DNS server instantly looks up [www.spotify.com])(https://www.spotify.com) in its database and replies with the correct public IP address belonging to Spotify's content delivery network (CDN).
- The Result: Armed with this numerical IP address, your phone can finally open a secure HTTPS connection to Spotify's servers and load your music library in milliseconds
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